

**CONSULTING SERVICES FOR PLANNING, SETTING-UP AND
OPERATIONALISING A REAL TIME FLOOD FORECASTING AND SPATIAL
DECISION SUPPORT SYSTEM FOR ADYAR, COOUM, KOSASTHALAYAR,
NAGARIAR, NANDHIYAR AND KOVALAM BASINS**

**CHENNAI FLOODS & REQUIREMENT OF
TAMIL NADU DISASTER RISK REDUCTION
AGENCY FOR FLOOD MANAGEMENT**

WORKSHOP – I

**Organised by Commissionerate of Revenue Administration & Disaster Management
and Tamil Nadu Urban Infrastructure Financial Services Limited**

**Hotel Crown Plaza,
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IMPACTS OF FLOODS IN CHENNAI AND ITS SUBURBS

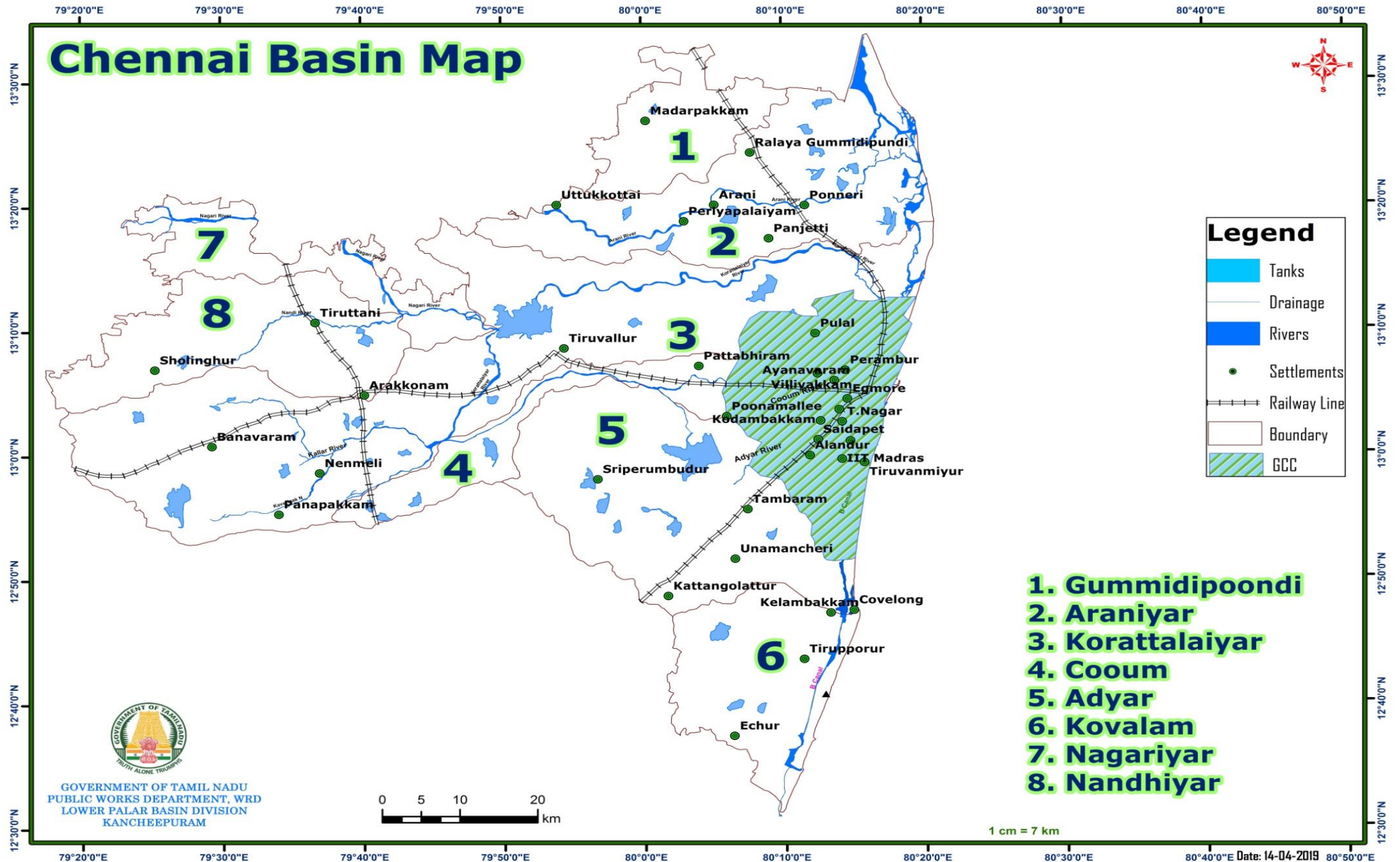
- ❖ Chennai City and its Suburbs had witnessed heavy damages to infrastructures and Loss of life during the years 1943, 1976, 1985, 1996, 2008, 2015 and 2020.
- ❖ The Coastal areas of the region overflowed and caused severe havoc on the urban areas of coastal region due to rapid urbanization and Climate change scenario in the past two decades.
- ❖ Government of India declared the floods in Chennai during the year 2015 as "CALAMITY OF SEVERE NATURE"



DRAINAGE SYSTEM OF CHENNAI BASIN

- ❖ **The Main Rivers Flowing in Chennai Basin are 1. Araniar, 2. Kosastahalaiyar, 3. Cooum and 4. Adayar.**
- ❖ **The Chennai Basin is further divided into 8 sub-basins according to drainage pattern namely 1. Gumidipoondi sub basin (an isolated water shed draining directly into sea through Pulicat lake), 2. Araniyar sub basin 3. Kosasthalayar sub basin, 4. Nandhiyar sub basin, 5. Nagariyar sub basin (Both the rivers are the tributaries of Kosasthalayar River), 6. Cooum sub basin, 7. Adayar sub basin and 8. Kovalam sub basin (an isolated watershed draining directly into sea through B'canal and Muttukadu back waters and Kovalam creek.**
- ❖ **Chennai basin covers an area of 6119 sq.km. which spread over in Chennai, Tiruvallur Districts and parts of Ranipet, Kancheepuram and Chengalpattu Districts.**
- ❖ **There are many isolated watersheds in between the rivers which are draining into sea through Buckingham Canal viz., Watersheds of Redhills Tank, Otteri Nullah, Captain Cotton Canal, Kodungayur Drain and Pallikkaranai.**
- ❖ **Greater Chennai and its Peri-urban areas are spread over an almost flat terrain along the coastal plains, having an average altitude of 2m to 6m above MSL.**
- ❖ **The entire surface runoff from Greater Chennai and its Peri-urban areas drain into Bay of Bengal only through the four outlets namely, Ennore creek on the north, Cooum and Adayar river mouths in the middle and Kovalam creek in the south.**
- ❖ **The integrity of the drainage network through Buckingham Canal, the Ennore backwaters, Muttukadu backwaters and Pallikaralai marsh land eco system are vital for flood mitigation.**

Chennai Basin Map



MAJOR CAUSES OF CHENNAI FLOODS / AREAS AFFECTED / STRUCTURAL AND NON-STRUCTURAL INTERVENTIONS REQUIRED FOR FLOOD RISK REDUCTION

❖ Cause 1

Change in Land use / Land cover resulted in obliteration of surplus courses of the Irrigation tanks which existed in the private agricultural lands which are now being converted into layouts, where open sheet flow prevailed earlier.

Areas affected

In North Chennai - areas developed along the surplus courses of Ambattur, Korattur, Kolathur, Madhavaram, Redhills, Kodungaiyur, Vysarpadi and Redhills tanks.

In Central Chennai – areas developed along the surplus courses of Mugapair tank, Porur tank and erstwhile Nungambakam and Mambalam tanks.

In south Chennai - areas developed along the surplus courses of Ottiyampakkam, Perumbakam, Madurapakkam, Guduvancheri, Urapakkam, Vandalur, Mannivakkam, Peerankaranai, Irumbuliyoore, Perunkalathur, Mudichur, Tambaram Periyaeri, Tambaram pudhuthangal, Selaiyur, Chitlapakkam, Nanmangalam and Pallikarnai tanks.

MAJOR CAUSES OF CHENNAI FLOODS / STRUCTURAL AND NON-STRUCTURAL INTERVENTIONS REQUIRED FOR FLOOD RISK REDUCTION – Contd...

Structural Interventions Required

- **Restoration of canal of the required size if adequate Government land is available.**
- **Construction of slender RCC retaining wall instead of earthen bund to increase the carrying capacity of the canal if the canal is narrowed due to urbanization.**
- **Construction of underground cut and cover macro drains below roads if no government land is available due to urbanization.**
- **At present most of the Minor Irrigation tanks are having uncontrolled weirs to discharge the surplus waters without having any regulating arrangements for creating buffer space to hold the inflows during extreme weather events. Hence, surplus regulators may be constructed to route the forecasted flood through the tank so that the peak flows are moderated.**

MAJOR CAUSES OF CHENNAI FLOODS / STRUCTURAL AND NON-STRUCTURAL INTERVENTIONS REQUIRED FOR FLOOD RISK REDUCTION – Contd...

Non - Structural Interventions Required

- “The Tamil Nadu Town and Country Planning act, 1971 (Tamil Nadu Act 35 of 1972)” provides for preparation of Master Plans, Zonal plans, Local Area Plans, Detailed Development Plan.
- Chennai Master Plan I came into force from 1975 for 1177 sq. km.
- Chennai Master Plan II came into force from 2008 for 1189 sq. km.
- Chennai Master Plan III is proposed for 2026 – 2046 for 6000 sq. km. (approx.)
- There is no guidelines in the above Act or in the Master Plans for the preservation and conservation of above lands falling in the flow path of surplus canals or in the flood plains along the rivers which were classified as patta dry / wet agricultural lands owned by private parties.
- Primarily such areas have to be surveyed and mapped in a scientific manner for floods of different return periods for which the ongoing RTFF & SDSS project will be most useful.
- So, amendment to rules in the above existing Act or enactment of new Act / rule to preserve and conserve such lands is needed before Chennai Master Plan III.

MAJOR CAUSES OF CHENNAI FLOODS / STRUCTURAL AND NON-STRUCTURAL INTERVENTIONS REQUIRED FOR FLOOD RISK REDUCTION – Contd...

❖ Cause 2

Insufficient carrying capacity of the rivers /urban drainages due to developmental constraints / sedimentation etc.,

Areas affected

In North Chennai - areas developed along the flood plains of Kosasthalaiyar river below Vallur Anicut in Napalayam, Manali Puthu Nagar, Sadayankuppam and Kargil nagar.

In Central Chennai – areas developed along the flood plains adjoining Cooum river in Vanagaram, Maduravoyal, Mugapair, Koyambedu, Anna Nagar and Aminjikarai.

In South Chennai - areas developed along the flood plains adjoining Adyar river in Mudichoor, Varadharajapuram, Thiruneermalai, Anakaputthur, Manapakkam, Nandambakkam, Jaffer Khanpet, Saidapet and Koturpuram.

MAJOR CAUSES OF CHENNAI FLOODS / STRUCTURAL AND NON-STRUCTURAL INTERVENTIONS REQUIRED FOR FLOOD RISK REDUCTION – Contd...

Structural Interventions Required

- **The stretches of rivers having inadequate width of waterway may be widened to the maximum width of the available Government poramboke lands with slender RCC retaining walls on both sides instead of earthen banks to increase the carrying capacity to desirable level .**
- **For micro and the arterial storm water drains falling into the river having its bed level well below the MFL of the river, the infrastructure for provision of shutters to prevent entering flood water into residential area with collection arrangements for pumping the storm water from the drain into the river may be explored.**
- **For the stretches of river having far less width to discharge the anticipated maximum flood discharge, feasibility studies for forming bypass canal or flood carrier canal off taking from the upper reaches to divert the in coming flood flows may be made.**

Non - Structural Interventions Required

- **Within Chennai Metropolitan area, the land use pattern of the private patta lands falling in the flow path of surplus canals or in the flood plains along the rivers have already been changed from agricultural use to other uses like residential / institutional / commercial / industrial use.**
- **Therefore, efficient system for Early Warning and effective plan for rescue and evacuation is considered essential to minimize risk due to floods in areas already developed in flood plains and along the flow path of surplus canals of tanks.**
- **The above phase for Disaster Management in Chennai City could be possible only when the on-going RTFF and SDSS with RTDAS is completed.**

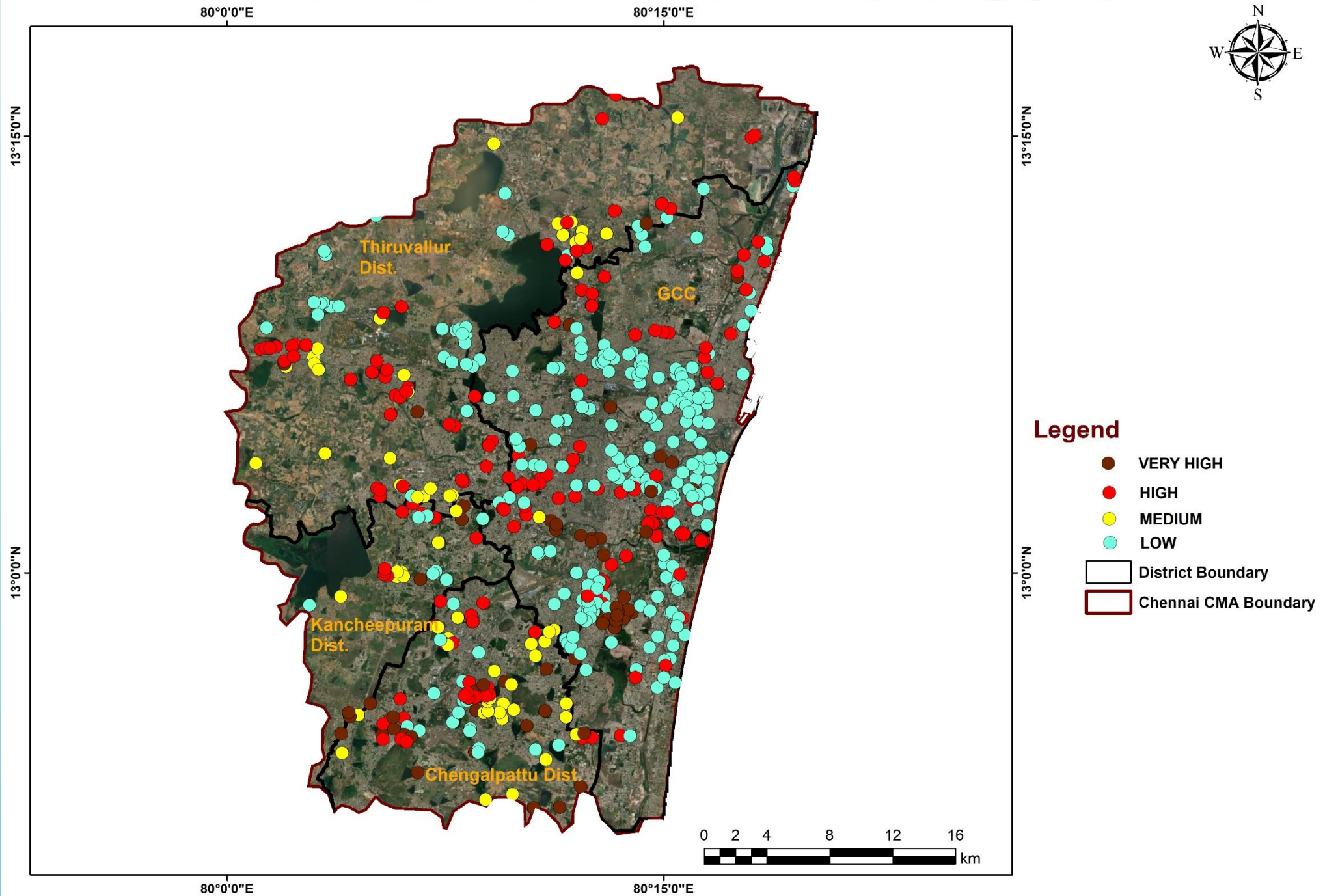
❖ Other Causes

- Reduction in the flow of water through the river mouth into sea due to increased tide level during the periods of cyclones and storm surges.
- Closure of river mouths caused by sedimentation due to littoral movements.
- Existence of naturally formed ecologically sensitive shoals within the creek / river mouth
- Discharging capacities of the discharging outlets are also under severe stress due to shrinking of the width of the creeks / River Mouths.

Structural Interventions Required

- Construction of properly oriented groynes / training walls on either side of the river mouth to minimize the quantum of sediments deposition in the creek and river mouth after conducting mathematical and proto type model studies.
- Annual program / plan for continuous dredging within the river mouth and creek to remove and dispose the minimal quantum deposition after the formation of groynes / training walls.

MAP SHOWING THE VULNERABLE LOCATIONS IN CMA AREA(As Per Legacy Data)



❖ Other Causes

- Sporadic and temporal variations in rainfall due to Climate Change scenario.
- Radical change in Land use / Land cover due to rapid urbanization leading to excess runoff.

Non structural interventions required

- Flood prone areas due to the above causes have to be identified in a scientific manner for floods of different return periods to have area specific Early Warning System, effective Rescue and Evacuation plan.
- Efficient model for Flood Forecast and Nowcast.

❖ Benefits of RTFF and SDSS for Chennai Basin area in GCC, Tiruvallur, Ranipet, Kancheepuram and Chengalpattu,

- Most of areas where above structural Interventions are required can be clearly identified in scientific manner using the DEM to be generated by SoI.
- The non structural interventions like location-specific Early Warning System, SOPs for releases from the Reservoirs and major water bodies and Effective Evacuation plan can be put into place.

Thank you

